PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Sleep disturbance in caregivers of individuals with Parkinsonism: a systematic review and meta-analysis.
AUTHORS	Sprajcer, Madeline; Owen, Patrick; Crowther, Meagan; Harper, Kirsty; Gupta, Charlotte; Ferguson, Sally; Gibson, RH; Vincent, Grace

VERSION 1 – REVIEW

REVIEWER	Partinen, Markku Vitalmed Research Center, Helsinki Sleep Clinic
REVIEW RETURNED	21-Mar-2022

GENERAL COMMENTS	This systematic review on caregiver's sleep is interesting. Methods are ok. Can you explain what the points of PSQI mean - as the PSQI may not be familiar to all readers? Can you provide information and/or discuss reasons for the poor sleep quality of the caregivers. Discuss a bit more limitations of your study.

REVIEWER	Gao, Chenlu Harvard Medical School, Sleep Medicine
REVIEW RETURNED	07-May-2022

GENERAL COMMENTS	The authors conducted a systematic review and meta-analysis and observed poor sleep quality in caregivers of patients with Parkinsonism. The topic is interesting and the methods are sound. I have some suggestions to help improve this manuscript: Major concerns: 1. I encourage the authors to carry out meta-regression analyses to investigate factors that influence the sleep quality of caregivers (e.g., age, gender). 2. Did the authors assess publication bias? If so, these results should be reported. 3. The "critical appraisal of included studies" section needs further elaborations. Could the authors discuss what are the weaknesses of past studies and what future researchers should do to improve the quality of research? E.g., Table 3 shows that all cross-sectional studies did well on items 2 and 3, but the majority did poorly on items 5 and 6.

	4. The authors should avoid causal language in the discussion, because all findings are correlational. E.g.: "A range of factors were identified as impacting sleep in this caregiving population, ..." Minor concerns: 1. In the abstract, it is better to clarify that 5.6 [4.8, 6.4] is for PSQI score. 2. Table 2 and Table 3 were referenced before Table 1. Please switch the order accordingly. 3. Table 1: Does the sample size column present the sample size of caregivers for patients with Parkinsonism or the total sample size for the study (i.e., including controls and other caregivers)? 4. Figure 2: please label the x-axis. 5. Page 9 line 56: "variance (e.g., SD) in this study was not reported". Variance is a statistic that equals SD squared. I think the authors meant "variability (e.g., SD)". 6. The authors used the phrase "delayed sleep latency" several times throughout the manuscript. Sleep latency is the amount of time it takes for an individual to fall asleep, so I'm not sure what "delayed sleep latency" means. Did you mean delayed sleep onset or increased sleep latency?

REVIEWER	Greenwell, Kate University of Southampton School of Psychology
REVIEW RETURNED	10-Jun-2022

GENERAL COMMENTS	Thank you for the opportunity to review this systematic review and meta-analysis of sleep disturbance in caregivers of adults with Parkinsonism. I thoroughly enjoyed reading this manuscript and commend the authors for shining light on an under-researched area. This manuscript was very well-written, the systematic review was carried out rigorously, and the discussion provided clear implications for clinical practice and future research. The identification of causal/predictive factors will be of particular value for informing the development of effective sleep interventions for this group. However, there are a few amendments I would like to see before it is accepted for publication. Major: 1. The review included studies that recruited “paid/unpaid/unprofessional caregivers”. I may just be misinterpreting what the authors mean by ‘paid’, but I would argue that a paid/formal carer (where caregiving is their job) is very different to an unpaid/informal carer (e.g. relative, friend). I would suggest that the authors exclude any studies that have recruited paid carers. If by ‘paid’ carer, the authors mean an informal/unprofessional carer that receives some kind of financial support (e.g. a carers allowance), then this just needs to be clarified in the manuscript. Minor: 2. Abstract: The results section was quite brief. It would be helpful to include the findings on factors associated with caregiver sleep.

	Perhaps you could shorten the objectives section if word count is an issue? 3. Aims in abstract and introduction: The main aim was quite broad and generally systematic reviews have very specific aims (they currently read more like scoping review aims). Please clarify what specific aspects of sleep disturbances you are researching (you are just interested in incidence of sleep disturbances and causal factors, not carers' sleep experiences/challenges or sleep interventions etc). In the first line of your discussion you give a much more specific review aim – perhaps you could use this (or something similar) instead? Also I think the aim relating to identifying causal factors should also be in the abstract. 4. Introduction: It would be helpful to clarify why it is important to do a review on the incidence/extent of sleep disturbances in caregivers of people with Parkinsonism and causal factors. How could your research potentially help improve the lives of caregivers and people with Parkinsonism? 5. Methods: Please can the authors clarify why they choose the age cut off for caregivers as 15+ years, as it seems a bit arbitrary. I'm also not sure I would class a 15 year old as an adult (as your review title states is your population of interest). Perhaps reconsider including the word 'adult' in the title (you don't refer to adults being your population of interest anywhere else). 6. Results: I wonder if 'causal factors' is the best term, given the results came from cross-sectional studies where causation cannot be inferred. Perhaps use a term like 'associated factors' or 'predictors' (if regression was used) instead? 7. Meta-analysis Results: Although the statistics are provided, it would be helpful to clarify in the text how the caregivers of people with Parkinsonism differed from healthy controls and other carers (e.g. were the sleep disturbances better or worse?). 8. Discussion: As per the PRISMA checklist, please discuss any limitations of the review process/methods used (not just the limitations of the included studies).

REVIEWER	Grainge, Matthew University of Nottingham, Division of Epidemiology and Public Health
REVIEW RETURNED	25-Aug-2022

GENERAL COMMENTS	In this systematic review, Sprajcer and colleagues report high levels of sleep disturbance in caregivers of adults with Parkinsonism. The review has been carried out to a high standard and has been reported clearly. I do not have too many specific comments. However, for someone with only limited knowledge of the topic area the review was difficult to follow as there was a lot of assumed knowledge about the outcome scales used. I would advise that a supplement table is produced clearly describing all of the scales used and how they are scored, even if this is at the expense of one of the existing tables in the supplement. Also, I was not clear about Sensitivity analysis 2 (in either Methods or Results). You omit studies that did not use PSQI, but only PSQI studies are included in your forest plot. Other minor comments: You have performed forward and backward citation searching from previously identified systematic reviews. What about forward and

	backward citation searching of studies included in the review. This would potentially pick up further missed studies. The final aim is to identify causal factors that may impact on sleep, but you have not really done this. The results paragraphs on this describe associations with sleep. To establish whether these are causal would be a whole new avenue to go down.

VERSION 1 – AUTHOR RESPONSE

REVIEWER #1 (PROF MARKKU PARTINEN)

Reviewer #1: This systematic review on caregiver's sleep is interesting. Methods are ok. Can you explain what the points of PSQI mean - as the PSQI may not be familiar to all readers?

Author response: Thank-you for sharing your time and expertise to review our manuscript, as well as your kind words. We have added the following to Methods > Statistical Analysis to further explain the scoring of the PSQI: *'The PSQI assesses seven separate sleep components (sleep quality, sleep onset latency, sleep duration, sleep efficiency, sleep disturbances, use of sleep medication and daytime dysfunction) and higher scores indicate poorer sleep.'* We address each of your other specific comments in details below.

Reviewer #1: Can you provide information and/or discuss reasons for the poor sleep quality of the caregivers.

Author response: Thank-you for this suggestion. We have added the following to Discussion > Paragraph 3: '*For caregivers of individuals with medical needs (including, but not limited to, individuals with Parkinsonism), sleep can be disrupted by a range of factors [70]. Aspects of the carer experience that impact sleep may differ depending on the individual needs of the care recipient, but include waking overnight to assist with tasks (e.g., visiting the bathroom), providing medical assistance (e.g., turning over in bed, administering medication), or helping to resettle their care recipient [70-72]. Additionally, sleep is often disrupted by the carer listening out for their care recipient in case they are needed [70, 73]. Worry and rumination about their care recipient is also a common cause of poor or disrupted sleep in carers [14, 74], as is the carer's own mental health [75].'*

Reviewer #1: Discuss a bit more limitations of your study.

Author response: Thank-you for this feedback. We have added the following to Discussion > Paragraph 9: *'In addition to the limitations of the body of evidence, there are several limitations of this review. First, the decision was made to include carers aged ≥ 15 years only which may limit the applicability of findings to younger caregivers. However, this decision was made in line with previous research undertaken on caring populations [90], and was chosen given the type of caring responsibilities addressed by this study (e.g., attending to medical needs, managing these needs in consultation with a healthcare team) would be less likely to be undertaken by children. For younger children who live with a family member who requires care, caring activities may instead involve, for example, a greater burden of household chores or childcare for younger siblings [91]. Additionally, studies published in languages other than English were excluded from this review, which may limit generalizability to non-English speaking countries. The present review also included studies that employed cross-sectional designs, which limits the capacity to infer causality. Furthermore, due to data paucity, meta-regressions investigating factors that influence sleep quality (e.g., age, gender) were not conducted.'*

REVIEWER #2 (DR CHENLU GAO)

Reviewer #2: The authors conducted a systematic review and meta-analysis and observed poor sleep quality in caregivers of patients with Parkinsonism. The topic is interesting and the methods are sound. I have some suggestions to help improve this manuscript:

Author response: Thank-you for sharing your time and expertise to review our manuscript, as well as your kind words. We address each of your specific comments in details below.

Major concerns

Reviewer #2: 1. I encourage the authors to carry out meta-regression analyses to investigate factors that influence the sleep quality of caregivers (e.g., age, gender).

Author response: Thank-you for this suggestion. We initially (per our PROSPERO registration: CRD42021274529) intended to conduct these meta-regressions pending data availability, however, none of our meta-analyses included 10 individual studies per Cochrane recommendations for meta-regression. To acknowledge this limitation, we have added the following to Discussion > Paragraph 7: *'Finally, due to data paucity, meta-regressions investigating factors that influence sleep quality (e.g., age, gender) were not conducted.'*

Reviewer #2: 2. Did the authors assess publication bias? If so, these results should be reported.

Author response: We apologise for the confusion. Publication bias was assessed as part of the GRADE approach. For clarity, we have added the following to Methods > Data Extraction: *'… which included assessment of publication bias via visual inspection of funnel plots.'*

Reviewer #2: 3. The "critical appraisal of included studies" section needs further elaborations. Could the authors discuss are what the weaknesses of past studies and what future researchers should do to improve the quality of research? E.g., Table 3 shows that all cross-sectional studies did well on items 2 and 3, but the majority did poorly on items 5 and 6.

Author response: Based on this feedback we have expanded on the 'Critical appraisal of included studies' section: '*Of the 18 included studies, four were case-control with quality assessment scores ranging from 3–10 (see Table 2). Of the case-control studies, one article was classed as "weak" [49], two were classed as "moderate" [29, 48], and one was classed as "strong" [50]. Common weaknesses of case-control studies included a lack of clarity as to whether case groups (i.e., carers of individuals with Parkinsonism) were comparable to, and matched appropriately with, control groups (aside from the caring role). Furthermore, few studies reported the length of time in the caregiving role, which resulted in an 'unclear' classification for the assessment item 'Was the exposure period of interest long enough to be meaningful?' for 75% of case-control studies. Case-control studies in this area should, in future, provide more information regarding both the matching of controls with cases, and the duration of caregiving.*

For the 14 cross-sectional studies included, the quality assessment scores ranged from 42.9% to 100% (see Table 3). Of the cross-sectional studies, three were classed as "weak" [36, 38, 42], five were classed as "moderate" [39, 44, 46, 47], and six were classed as "strong" [30, 37, 40, 41, 43, 45]. Of note, all cross-sectional studies were classified as 'N/A' for the item 'Were objective, standard criteria used for measurement of the condition?', given that there are no existing standardised criteria used to identify individuals as carers. Additionally, all but one cross-sectional study did not identify or manage

confounding factors. Future cross-sectional research in this area would benefit for the identification of potentially confounding factors, given that carers may have, for example, personal health conditions [63] or other traits that may impact their experience [64].'

Reviewer #2: 4. The authors should avoid causal language in the discussion, because all findings are correlational. E.g.: "A range of factors were identified as impacting sleep in this caregiving population, ..."

Author response: Thank-you for his recommendation. We have removed all instances of causal language throughout the Discussion. We have also added the following to the limitations section of our discussion: *'The present review also included studies that employed cross-sectional designs, which limits the capacity to infer causality.'*

Minor concerns

Reviewer #2: 1. In the abstract, it is better to clarify that 5.6 [4.8, 6.4] is for PSQI score.

Author response: Thank-you for this suggestion. We have amended this statement in Abstract > Results to the following: '… (mean [95%CI]: 5.6 [4.8, 6.4] points on the Pittsburgh Sleep Quality Index) …'

Reviewer #2: 2. Table 2 and Table 3 were referenced before Table 1. Please switch the order accordingly.

Author response: This has been updated.

Reviewer #2: 3. Table 1: Does the sample size column present the sample size of caregivers for patients with Parkinsonism or the total sample size for the study (i.e., including controls and other caregivers)?

Author response: This table represents the number of caregivers for individuals with Parkinsonism. We have updated the column header to reflect this in Table 1.

Reviewer #2: 4. Figure 2: please label the x-axis.

Author response: Thank-you for this suggestion. We have added the following label: *'Points on the Pittsburgh Sleep Quality Index'*.

Reviewer #2: 5. Page 9 line 56: "variance (e.g., SD) in this study was not reported". Variance is a statistic that equals SD squared. I think the authors meant "variability (e.g., SD)".

Author response: Thank-you for this comment. We have amended this to *'spread' to best represent the type of summary statistic*.

Reviewer #2: 6. The authors used the phrase "delayed sleep latency" several times throughout the manuscript. Sleep latency is the amount of time it takes for an individual to fall asleep, so I'm not sure what "delayed sleep latency" means. Did you mean delayed sleep onset or increased sleep latency?

Author response: We would like to thank the reviewer for picking up on this error in terminology. We have replaced ‘delayed sleep latency’ with ‘increased sleep latency’ throughout.

REVIEWER #3 (DR KATE GREENWELL)

Reviewer #3: Thank you for the opportunity to review this systematic review and meta-analysis of sleep disturbance in caregivers of adults with Parkinsonism. I thoroughly enjoyed reading this manuscript and commend the authors for shining light on an under-researched area. This manuscript was very well-written, the systematic review was carried out rigorously, and the discussion provided clear implications for clinical practice and future research. The identification of causal/predictive factors will be of particular value for informing the development of effective sleep interventions for this group. However, there are a few amendments I would like to see before it is accepted for publication.

Author response: Thank-you for sharing your time and expertise to review our manuscript, as well as your kind words. We address each of your specific comments in details below.

Major

Reviewer #3: 1. The review included studies that recruited “paid/unpaid/unprofessional caregivers”. I may just be misinterpreting what the authors mean by ‘paid’, but I would argue that a paid/formal carer (where caregiving is their job) is very different to an unpaid/informal carer (e.g. relative, friend). I would suggest that the authors exclude any studies that have recruited paid carers. If by ‘paid’ carer, the authors mean an informal/unprofessional carer that receives some kind of financial support (e.g. a carers allowance), then this just needs to be clarified in the manuscript.

Author response: We agree that the description of caregivers as ‘paid’ was unclear in the manuscript. We have updated this section to reflect the use of the work ‘paid’ to refer to individuals who receive some form of government or other allowance as compensation for their time, rather than employment. Changes can be seen in Methods > Eligibility criteria: *‘Inclusion criteria for this review were: (a) population of paid (in the form of a carers allowance or similar stipend only) or unpaid, non-professional caregivers (≥15 years of age) of a person with any form of Parkinsonism (e.g., Parkinson’s disease, drug-induced Parkinsonism, vascular Parkinsonism, multiple systems atrophy, progressive supranuclear palsy, and corticobasal degeneration)’*

Minor

Reviewer #3: 2. Abstract: The results section was quite brief. It would be helpful to include the findings on factors associated with caregiver sleep. Perhaps you could shorten the objectives section if word count is an issue?

Author response: While we would like to be able to definitively identify which factors were associated with poor sleep in this caregiving population, the factors identified were largely speculative in nature. As such we do not believe the data supporting potential factors have the strength to be included in the abstract.

Reviewer #3: 3. Aims in abstract and introduction: The main aim was quite broad and generally systematic reviews have very specific aims (they currently read more like scoping review aims). Please clarify what specific aspects of sleep disturbances you are researching (you are just interested in incidence of sleep disturbances and causal factors, not carers' sleep experiences/challenges or sleep interventions etc). In the first line of your discussion you give a much more specific review aim – perhaps you could use this (or something similar) instead? Also I think the aim relating to identifying causal factors should also be in the abstract.

Author response: We agree that the specific aim identified in the discussion should be consistent throughout our review. As such, we have updated the abstract and introduction with this aim: '*The aim of this systematic review and meta-analysis was to investigate whether providing care to an individual with Parkinsonism was associated with disturbed caregiver sleep, and to identify associated factors that may contribute to disturbed sleep in this population.'*

Reviewer #3: 4. Introduction: It would be helpful to clarify why it is important to do a review on the incidence/extent of sleep disturbances in caregivers of people with Parkinsonism and causal factors. How could your research potentially help improve the lives of caregivers and people with Parkinsonism?

Author response: We have added the following clarification in our introduction regarding the potential benefit of undertaking this review (Introduction > Paragraph 3): '*By identifying the extent to which caring responsibilities result in disturbed sleep in this population, and what factors may be associated with poor sleep, we will be able to provide evidence to underpin the provision of additional, targeted support to caregivers of individuals with Parkinsonism.'*

Reviewer #3: 5. Methods: Please can the authors clarify why they choose the age cut off for caregivers as 15+ years, as it seems a bit arbitrary. I'm also not sure I would class a 15 year old as an adult (as your review title states is your population of interest). Perhaps reconsider including the word 'adult' in the title (you don't refer to adults being your population of interest anywhere else).

Author response: Thank you for this feedback. We have updated the title to omit the word 'adults', which we have replaced with 'individuals'. Furthermore, we have clarified our decision to include only individuals aged 15 and over in this study, and have included it as a limitation of our review, Discussion > Paragraph 9: *'In addition to the limitations of the body of evidence, there are several limitations of this review. First, the decision was made to include carers aged ≥ 15 years only which may limit the applicability of findings to younger caregivers. However, this decision was made in line with previous research undertaken on caring populations [90], and was chosen given the type of caring responsibilities addressed by this study (e.g., attending to medical needs, managing these needs in consultation with a healthcare team) would be less likely to be undertaken by children. For younger children who live with a family member who requires care, caring activities may instead involve, for example, a greater burden of household chores or childcare for younger siblings [91].'*

Reviewer #3: 6. Results: I wonder if 'causal factors' is the best term, given the results came from cross-sectional studies where causation cannot be inferred. Perhaps use a term like 'associated factors' or 'predictors' (if regression was used) instead?

Author response: We agree with this comment and have updated the manuscript with the term 'associated factors' throughout. We have also added the following to the limitations section of our discussion: *'The present review also included studies that employed cross-sectional designs, which limits the capacity to infer causality.'*

Reviewer #3: 7. Meta-analysis Results: Although the statistics are provided, it would be helpful to clarify in the text how the caregivers of people with Parkinsonism differed from healthy controls and other carers (e.g. were the sleep disturbances better or worse?).

Author response: Thank-you for this suggestion. We have amended the statement within Results > Meta-analysis of Caregiver Sleep to: *'Caregivers of individuals with Parkinsonism had lower sleep quality compared to …'*.

Reviewer #3: 8. Discussion: As per the PRISMA checklist, please discuss any limitations of the review process/methods used (not just the limitations of the included studies).

Author response: Thank-you for flagging this omission. We have added the following limitations section regarding the conduct of our study: We have added the following to Discussion > Paragraph 9: *'In addition to the limitations of the body of evidence, there are several limitations of this review. First, the decision was made to include carers aged ≥ 15 years only which may limit the applicability of findings to younger caregivers. However, this decision was made in line with previous research undertaken on caring populations [90], and was chosen given the type of caring responsibilities addressed by this study (e.g., attending to medical needs, managing these needs in consultation with a healthcare team) would be less likely to be undertaken by children. For younger children who live with a family member who requires care, caring activities may instead involve, for example, a greater burden of household chores or childcare for younger siblings [91]. Additionally, studies published in languages other than English were excluded from this review, which may limit generalizability to non-English speaking countries. The present review also included studies that employed cross-sectional designs, which limits the capacity to infer causality. Furthermore, due to data paucity, meta-regressions investigating factors that influence sleep quality (e.g., age, gender) were not conducted.'*

REVIEWER #4 (DR MATTHEW GRAINGE)

Reviewer #4: In this systematic review, Sprajcer and colleagues report high levels of sleep disturbance in caregivers of adults with Parkinsonism. The review has been carried out to a high standard and has been reported clearly. I do not have too many specific comments.

Author response: Thank-you for sharing your time and expertise to review our manuscript, as well as your kind words. We address each of your specific comments in details below.

Reviewer #4: However, for someone with only limited knowledge of the topic area the review was difficult to follow as there was a lot of assumed knowledge about the outcome scales used. I would advise that a supplement table is produced clearly describing all of the scales used and how they are scored, even if this is at the expense of one of the existing tables in the supplement. Also, I was not clear about Sensitivity analysis 2 (in either Methods or Results). You omit studies that did not use PSQI, but only PSQI studies are included in your forest plot.

Author response: Based on this comment, and feedback from another reviewer, we have included more detailed information about the key outcome measure (PSQI) in the body of the manuscript (Methods > Statistical Analysis > Paragraph 2): '*The PSQI assesses seven separate sleep components (sleep quality, sleep onset latency, sleep duration, sleep efficiency, sleep disturbances, use of sleep medication and daytime dysfunction) and higher scores indicate poorer sleep [31].'* However, given that all other scales were used by few studies (n = 1 for all other scales), we have included citations for each scale that can be accessed by the reader rather than including additional supplementary materials. For Sensitivity Analysis 2, we apologise for the confusion, this analysis omitted studies that were transposed to a 21-point scale given PSQI was the most commonly implemented tool. For clarity, we have added the following to Results > Meta-analysis of Caregiver Sleep: *'… data (i.e., omission of other tools transposed to a 21-point scale) …'*.

Other minor comments

Reviewer #4: You have performed forward and backward citation searching from previously identified systematic reviews. What about forward and backward citation searching of studies included in the review. This would potentially pick up further missed studies.

Author response: We apologise for the confusion. We did implement forward and backwards citation tracking of included studies. We have amended the statement within Methods > Information Sources to the following: *'Backward and forward citation tracking was conducted on the included articles.'*

Reviewer #4: The final aim is to identify causal factors that may impact on sleep, but you have not really done this. The results paragraphs on this describe associations with sleep. To establish whether these are causal would be a whole new avenue to go down.

Author response: We have clarified that our aim was to identify associated factors rather than causal, and have updated the terminology as such throughout the manuscript. We have also added the following to the limitations section of our discussion: *'The present review also included studies that employed cross-sectional designs, which limits the capacity to infer causality.'*

VERSION 2 – REVIEW

REVIEWER	Greenwell, Kate University of Southampton School of Psychology
REVIEW RETURNED	12-Oct-2022

GENERAL COMMENTS	I am satisfied that the authors have thoroughly addressed all of my comments and made appropriate revisions.

REVIEWER	Grainge, Matthew University of Nottingham, Division of Epidemiology and Public Health
REVIEW RETURNED	20-Oct-2022

GENERAL COMMENTS	Thank you for the efforts made in revising this paper. My concerns have been addressed however, please considering changing your last edit to "The present review also included studies that employed observational designs, which limits the capacity to infer causality". This is because causality cannot be inferred from any observational design including cohort.